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Palaeomagnetism of Cretaceous carbonates from the northwestern part of the Dinaric fold belt

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Abstract

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From the island of Cres and the fold belt north of autochthonous Istria 260 light-grey Cretaceous limestone samples were collected at 25 localities. Stepwise thermal demagnetization was employed to clean the remanence up to max. 525°C. About two-thirds of the samples were successfully cleaned and the locality means were calculated from the characteristic remanences. The locality means significantly deviate from the present field direction both before and after tectonic correction: the mean inclination is about 15° lower than the present inclination and the declination is rotated counterclockwise, i.e. it basically fits the pattern so far observed in the Periadriatic region.

Unfolding barely influences the scatter of the directions (Cres: $D = 308°$, $I = 45°$, $k = 26$, $\alpha_{95} = 13.3°$, before tilt correction, and $D = 330°$, $I = 48°$, $k = 33$ $\alpha_{95} = 11.9°$ after tilt correction; fold belt north of autochthonous Istria: $D = 309°$, $I = 48°$, $k = 17$, $\alpha_{95} = 11.9°$, before tilt correction, and $D = 336°$, $I = 41°$, $k = 13$, $\alpha_{95} = 14.0°$, after tilt correction. In view of the inconclusive fold test, the magnetization cannot be dated precisely and thus does not yield sufficient information concerning possible relative movements between autochthonous Istria–Gargano and the fold belt studied. Nevertheless, the counterclockwise rotated declinations of the area studied are in contrast with the clockwise rotated declinations of the Ionian Zone of Western Greece. The large angular difference in declination between two parts of the Dalmatian–Ionian Zone with similar tectonic trends casts doubt on the continuity of the fold belt.

Introduction

In recent years major advances have been made in Mediterranean palaeomagnetism. The development of highly sensitive rock magnetometers allows the use of the weakly magnetized limestones and dolomites which are widespread in the Periadriatic region, the Alps and the Carpatho-Balkanids. The enlarged and more precise data base has contributed to the progress, but other factors, such as recognition of the key importance of the autochthonous areas and the concentration of efforts to within certain areas in order to construct apparent polar wander curves, have been equally important. Now it is possible to interpret the palaeomagnetic results obtained from thrust sheets of weakly magnetized carbonates in terms of relative movements not only with respect to the major plates but to other, more or less stable tectonic units within the Mediterranean.

On the evidence of the existing geological data, the northwestern part of the Dinarides—the object of the present palaeomagnetic study—is parautochthonous. The samples were collected from different overthrust sheets belonging to the Dalmatian–Herzegovinian Zone (Dimitrijevic, 1982), both on the mainland north of autochthonous peninsular Istria and on the island of Cres.

The study was designed to provide palaeo-

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TABLE 1

Palaeomagnetic results

Sampling locality	Age	N/N_0	D^0	I^0	k	α_{95}^0	D_c^0	I_c^0	k	α_{95}^0	Cleaning
Cres, Cretaceous limestones and marls											
1. Punta Kriza	Barremian–Aptian	11/13	322	35	29	8.6	324	47	56	6.1	300°–500°C
2. Krcina	Barremian–Aptian	9/9	324	63	135	4.4	348	50	135	4.4	150°–500°C
3. Osor	Cenomanian–Turonian	12/12	320	53	52	7.2	346	41	52	7.2	100°–400°C
4. Valun	Cenomanian–Turonian	7/7	293	31	30	11.0	312	58	30	11.0	100 Oe 150°–500°
5. Valun	Cenomanian–Turonian	11/12	306	38	36	7.7	306	38	20	10.6	150°–500°C
6. Predoscica	Cenomanian–Turonian	7/7	292	45	31	10.9	341	46	31	10.9	400°C
Early Cretaceous mean (2 loc.)		20/22	324	49	–	–	336	49	–	–	
Late Cretaceous mean (4 loc.)		37/38	302	42	38	15.2	327	47	24	19.2	
Cretaceous mean (6 loc.)		57/60	308	45	26	13.3	330	48	33	11.9	
Mean, loc. 1, 4, 5, 6			303	37	41	14.9	321	48	37	15.4	
Mean, loc. 1, 4, 6, 7, 9, 13			300	44	18	10.4	323	52	53	9.3	
Mainland north and northeast of autochtonous Istria, Cretaceous limestones and marls											
7. Mune	Barremian	6/6	304	42	28	12.8	325	45	28	12.8	150°–500°C
8. Mune	Barremian–Aptian	6/6	312	46	46	10.4	340	29	46	10.4	150°–500°C
9. Vodice	Barremian–Aptian	9/9	332	60	38	9.1	317	48	38	9.1	150°–525°C
10. Poljane	Valanginian–Barremian	6/7	305	36	23	14.4	332	38	23	14.4	200°–525°C
11. Icici	Barremian–Albian	6/6	323	25	–	15.6	344	45	–	15.6	400°–500°C
12. Vodice	Turonian	6/6	293	50	26	13.3	332	44	26	13.3	150°–500°C
13. Racice	Turonian	4/6	85	−42	39	14.8	135	−68	39	14.8	150°–300°C
14. Brezovica	Turonian–Senonian	16/18	318	51	11	11.5	0	42	11	11.5	100 Oe 400°C
15. Golac	Turonian–Senonian	11/18	306	42	6	21.2	305	26	6	21.2	150°–400°C
16. Hrusica	Senonian	6/7	4	64	8	25.1	15	10	8	25.1	150 Oe 200°C
Early Cretaceous mean (5 loc.)		33/34	314	42	28	14.7	332	41	54	10.5	
Late Cretaceous mean (5 loc.)		43/55	303	54	13	22.0	341	41	7	32.3	
Cretaceous mean (10 loc.)		76/89	309	48	17	11.9	336	41	13	14.0	
Late Cretaceous mean, 16 excluded (4 loc.)			295	48	25	18.9	328	47	11	28.9	
Cretaceous mean, 16 excluded (9 loc.)			306	45	24	10.7	331	44	22	11.1	
Loc. 7, 9–13			303	44	45	17.7	329	48	43	10.3	

magnetic directions from mainly Cretaceous sediments, because an earlier work on similar rocks in autochthonous position indicated (Márton and Veljovic, 1983) that neither the Jurassic nor the Eocene limestones and sandstones yielded consistent directions.

The aims of this palaeomagnetic study were (1) to determine whether or not there has been a bulk rotation of the Dalmatian–Herzegovinian Zone relative to its stable foreland, i.e. autochthonous Istria and Gargano, (2) to find out if the nappe transport resulted in a spread of the palaeomagnetic declinations and (3) to resolve whether the Dalmatian and Ionian zones (Celet, 1977) form a continuous chain of folds.

Sampling and laboratory treatment

With two exceptions—one is a mid-Jurassic white limestone, and the other an Eocene dark grey sandstone (Fig. 1, loc. 19 and 25), the rocks collected were white to dark grey limestones and dolomites of Cretaceous age.

The initial remanence was measured on sister specimens with a JR-3 Spinner in Beograd and JR-4 Spinners in Budapest, followed by AF (alternating field) cleaning in Beograd up to 10 mT and progressive AF plus thermal demagnetization in Budapest. Changes in magnetic susceptibility during heating were monitored with a KLY-2 susceptibility bridge. All cores were heated at 150°C to remove any signal residing in hydrous iron oxides, and based on the behaviour of pilot specimens (e.g. Fig. 2) higher cleaning temperatures were subsequently selected, for each locality.

In spite of the weak initial remanences, characteristic magnetizations for 16 localities out of the 25 (260 samples) could be isolated, since the NRM cleaned readily and the usually diamagnetic susceptibilities remained reasonably constant during heating.

Palaeomagnetic results

Thermal demagnetization is obviously superior to AF for removing overprint in sediments. Most of the locality means in Table 1 are therefore based on thermally cleaned NRM directions. The

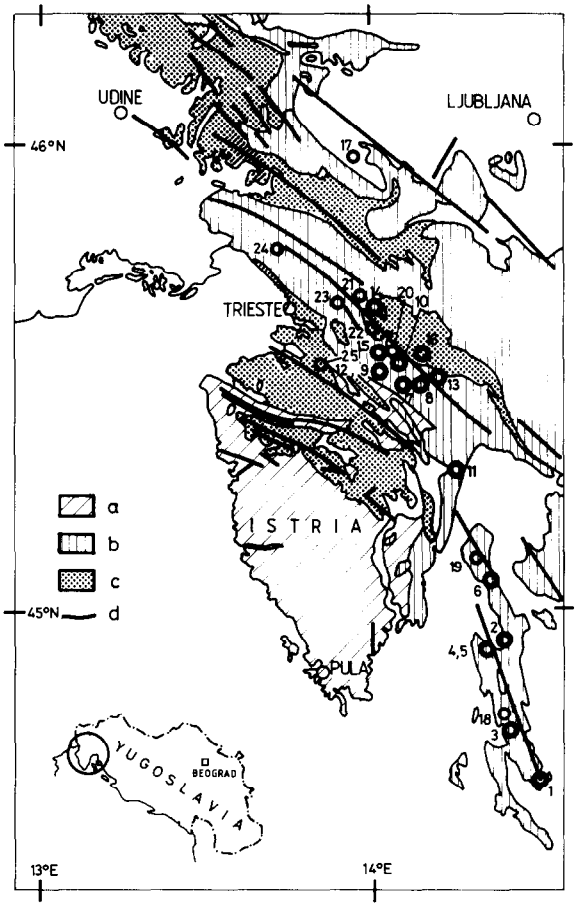


Fig. 1. Geological sketch map after the 1 : 1,500,000 geological map of Italy (edited by the Consiglio Nazionale delle Richerche, Rome, 1980) with the palaeomagnetic sampling localities (circles). Numbers *1–16* refer to Table 1., localities with higher number were unsuccessful: *17* is a mid-Jurassic white limestone, *18–24*–Cretaceous carbonates, *25*–Eocene dark grey sandstone. Key to the geological map: *a* = outer zone of the carbonate platform-basin system, *b* = Austroalpine Permo-Mesozoic, *c* = flysch, *d* = orientation of fold axes.

remanence, after removing the overprint (e.g., Fig. 2E 450°C, and Fig. 2G 150°C) became very weak. The characteristic magnetization for each specimen was calculated therefore as the average of two or more directions observed at subsequent cleaning steps above the temperature of the practical removal of the overprint component. For localities 4, 14 and 16, AF and thermally cleaned NRM is combined. The reason is that some members of these groups lost their NRM at the first step of the heating (150°C), but were reasonably stable on AF cleaning and yielded directions simi-

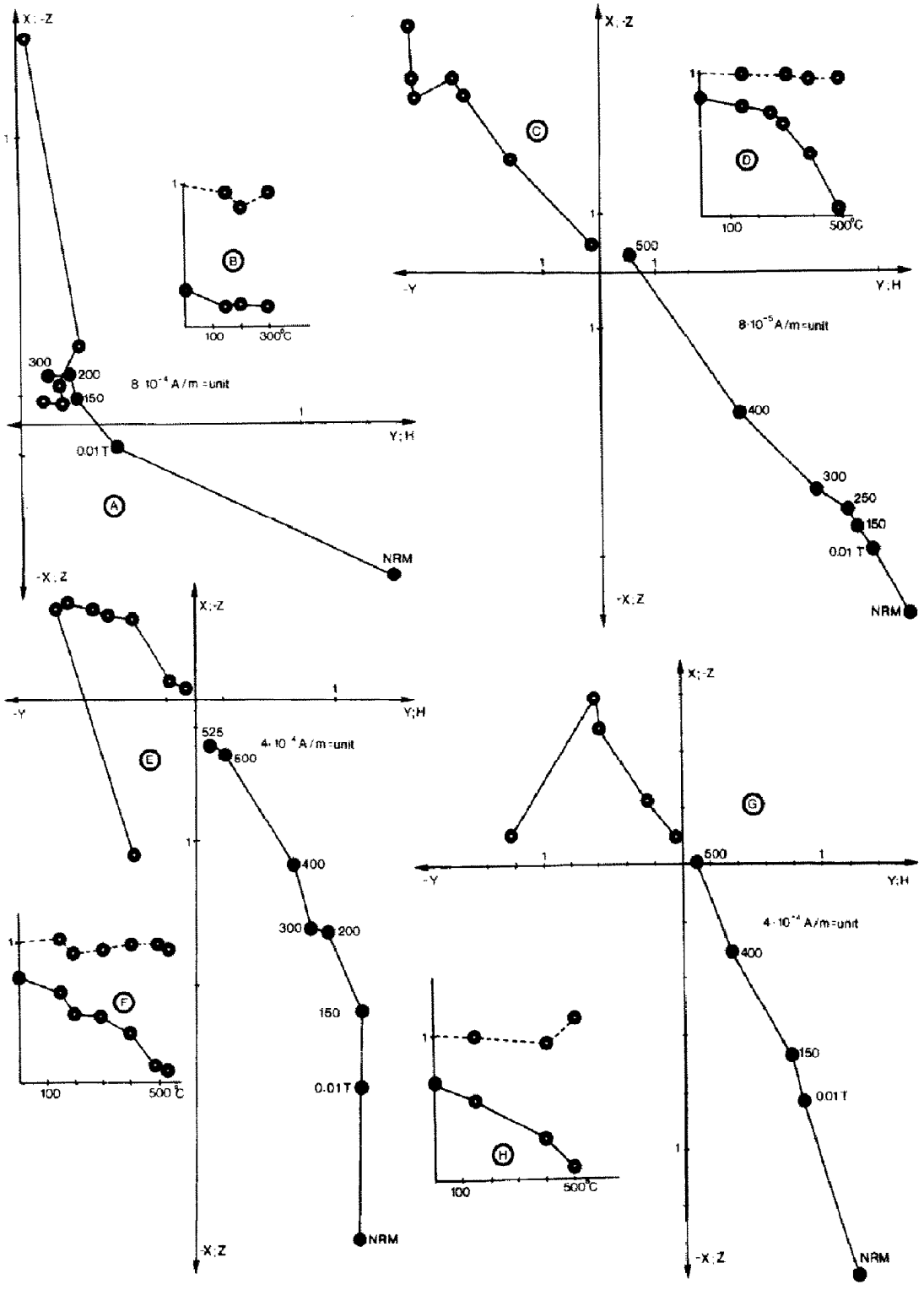


Fig. 2. Demagnetization diagrams in orthogonal projection (A, C, E, G), normalized intensity and susceptibility curves (B, D, F, H) of four characteristic specimens from Cretaceous carbonates. Open and solid dots on vector plots indicate declination and inclination, respectively; numbers following the 0.01 T demagnetization step refer to temperature in °C. Intensity (solid dots) is normalized to the intensity measured after the 0.01 T demagnetization step. Susceptibility (open dots) is normalized to the initial value.

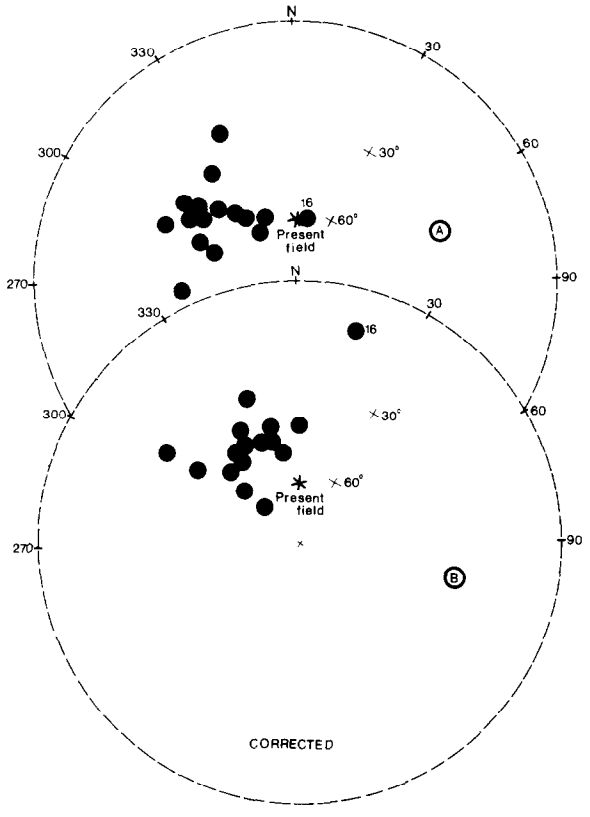


Fig. 3. All site means of Table 1 plotted on stereographic projection, (A) without bedding correction, and (B) corrected for local tilt. The site mean directions are represented by dots and all plot on the lower hemisphere.

lar to those of the thermally cleaned specimens of the same group.

Although weak, several carbonate varieties proved to be excellent recorders of the ancient magnetic field. Reasonably well defined directions were obtained for 16 localities (Table 1). The uncorrected mean of locality 16 is close to the present Earth's field direction while the others depart counterclockwise (Fig. 3A). The same locality mean significantly differs from the cluster of the directions on tilt correction (Fig. 3B); therefore it is disregarded for tectonic interpretation.

Discussion

As can be seen from Fig. 3 and Table 1, the palaeomagnetic directions exhibit the same amount of scatter before and after simple bedding tilt correction. The scatter of the tilt-corrected NRM does not exceed the limit usual for an autochthonous region where the rocks are weakly magnetized and not of the same geological stage.

In Fig. 3B, the spread in declination is similar to that of the inclination, which suggests that neither plunging fold axes (simple tilt correction was applied) nor nappe rotations can account for the scatter.

Orogenic trend is near-constant at the sampling localities with meaningful directions (Fig. 1, larger circles), apart from a roughly 30° systematic difference between Cres island and the mainland.

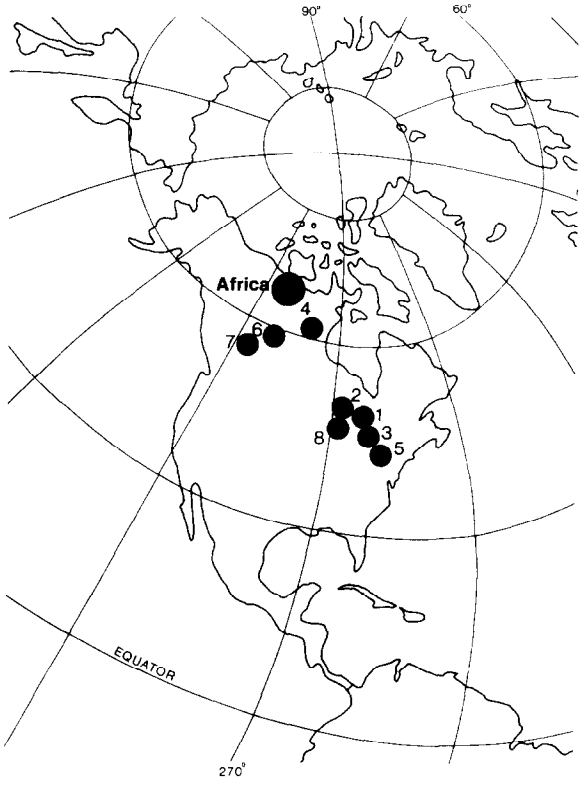


Fig. 4. Comparison between the palaeomagnetic poles from Cres (*3*–uncorrected, *4*–corrected for tilts), and the mainland north of autochtonous Istria (*5*–uncorrected, *6*–corrected for tilts) with Cretaceous mean poles from autochtonous Istria (*1*–uncorrected, *2*–corrected for tilt), Gargano (*8*–corrected for tilt), the Southern Alps (*7*–corrected for tilts) and Africa. *References*: pole *1* and *2*–Márton and Veljovic, (1983); poles *3*–*6*—present study; pole *7*–Channell and Tarling (1975), VandenBerg and Wonders, (1980); pole *8*–VandenBerg (1983); pole for Africa—calculated from data compiled by Schult et al. (1981).



Fig. 5. Orientation of the fold axes and palaeomagnetic mean directions for the Central Mediterranean. Fold axes after the tectonic map of Europe (Carte Tectonique Internationale de l'Europe 1 : 2,500,000, 1964, Moscow). Palaeomagnetic mean directions are as follows: Cres (*C*): Cretaceous mean–present study; Dinaric fold belt (*D*) north of autochtonous Istria–Cretaceous mean, present study; autochtonous Istria (*I*): Cretaceous mean–Márton and Veljovic (1983); Gargano (*G*) Cretaceous mean–VandenBerg (1983); Umbrian Apennines (*UA*): Cretaceous mean–Channell and Tarling (1975), VandenBerg and Wonders (1980); Transdanubian Central Mountains (*TC*): Cretaceous mean–Márton and Márton (1983); Epiros, Western Greece (*E*): Eocene mean (corrected for tilt plus respective local fold axes)–Horner and Freeman (1983).

This difference is not reflected in the overal mean for Cres and the mainland respectively (Table 1).

In view of this, the grouping may not be improved further, i.e. the results of an inconclusive fold test are to be interpreted in relation to other areas of southern Tethyan origin.

Cretaceous rocks from two autochthonous areas close to the area of the present study were earlier investigated palaeomagnetically: Istria (Márton and Veljovic, 1983) and Gargano (Channell and Tarling, 1975; VandenBerg, 1983).

The overall mean declination based on sufficiently cleaned Cretaceous samples from both Istria and Gargano is counterclockwise rotated and the angle is about 30° higher than it would be in an African reference system.

The mean declination for Cres and the fold belt north of autochthonous Istria before tilt correction is similar to that observed for the autochthon. The tilt-corrected directions, however, imply an

about 30° clockwise rotation of the northwestern part of the Dalmatian Zone with respect to Istria–Gargano, i.e. they practically agree with directions of presumably the same age from Africa and the Southern Alps (Fig. 4).

Clearly, the age of the magnetization is critical for the correct interpretation of the relative position of the autochthon and the studied parautochthon.

Concerning the autochthon, the magnitude of the declination rotation is well-established, because the correction for local tilt—since the strata are in sub-horizontal position—has practically no influence on the overall mean. For the same reason the age of the weak, therefore only moderately grouping NRM, could not be proved by fold test. Nevertheless, the original magnetization is more likely to persist in weakly deformed rather than in strongly deformed sediments.

The most probable model therefore is that Istria–Gargano and the area of the present study belong together, but the palaeomagnetic direction for the latter (assumed as syn- or postfolding NRM on Cretaceous rocks) reflects a younger stage of the movement with respect to the major plates than does the direction for the autochthonous (accepted as prefolding magnetization).

Another, but less likely assumption is that the NRM is of syn- or postfolding age for both the autochthon and the parautochthon. In this case we may move them together after the Eocene (age of the folding in the Dalmatian Zone) but have to conclude that the rotation with respect to Africa is larger than thought previously.

In case the NRM should be primary in the autochthonous as well as in the parautochthonous areas, we have to find a solution to a situation similar to the puzzling relationship between the palaeomagnetic declinations for the Southern Alps and the Umbrian Apennines (VandenBerg and Wonders, 1980), i.e. it would mean a bulk clockwise rotation of the fold belt relative to the autochthonous since the Cretaceous, corresponding to the difference between the orogenic trend in Istria (E–W) and the northwestern part of the Dalmatian Zone (SE–NW) respectively).

This kind of apparent correlation should, however, be treated with caution. It has been shown (Márton, 1984) that correlation between fold orientation and palaeomagnetic declination is generally lacking in the central Mediterranean: elements of southern Tethyan origin, like Gargano, Istria, the Umbrian Apennines and the Transdanubian Central Mountains have different structural trends yet the palaeomagnetic directions agree fairly well (Fig. 5). The same applies to the Southern Alps and the area of the present study. On the contrary, the orientation of the fold axes is the same in the area of the present study as in Epiros (Western Greece), yet the declinations are rotated counterclockwise in the first, and clockwise in the second area (Fig. 5).

The contrast between the uniformly oriented fold axes and the widely different palaeomagnetic declinations is so large that it cannot be entirely attributed to age difference in the magnetization. It is more likely that we are dealing with two units with a different rotational history, with the Scutari–Pec line (Fig. 5) as a probable suture between them, i.e. the Dalmatian–Ionian Zone could not have existed until after the late Eocene.

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